IoT-Based Environmental Comfort and Air Quality Monitoring for The Helix Residences

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Declaration of Originality and Ethical Use of AI

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I declare that I have not engaged in any form of academic misconduct, including plagiarism, collusion, fabrication, or falsification of data. All use of others' work has been appropriately cited.

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Abstract / Executive Summary

This report presents a simulated Internet of Things system for The Helix Residences in the Helix Park smart city scenario, focusing on environmental comfort and air quality in communal residential spaces. The aim of the project was to design a privacy-aware monitoring solution that could help facilities staff move from reactive complaint handling towards more proactive, evidence-based decision-making. The system combines a Wokwi-based ESP32 sensing node with a Python desktop dashboard that simulates five communal zones, applies threshold logic, generates alerts, stores logs, visualises trends, and produces operational recommendations. The approach used was design-based and simulation-led, with a modular software structure covering data generation, rule processing, alert management, analytics, logging, and graphical presentation. The project was evaluated through its ability to generate realistic zone-specific behaviour, support historical analysis, and expose meaningful environmental anomalies such as ventilation failure and sustained air-quality decline. The final artefact demonstrates that a relatively lightweight IoT architecture can still address a real smart-building problem in a technically coherent way, while also respecting residential privacy constraints by limiting monitoring to communal environmental conditions only. The report concludes that the solution is strong as a simulated artefact, but a real deployment would require authenticated devices, encrypted communications, richer validation, and wider integration with smart-city infrastructure.

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1. Introduction

The aim of this project was to design and implement a simulated IoT monitoring system for The Helix Residences, one of the organisations in the Helix Park smart city scenario. The chosen operational domain was environmental comfort and air quality, with particular attention to communal spaces such as the lobby, gym, co-working suite, corridor, and rooftop lounge. The central problem is that poor environmental conditions in these spaces may persist for too long before facilities staff notice them, leading to complaints, avoidable discomfort, and weak evidence for maintenance decisions.

The report has three main objectives. First, it explains why this problem is important in the residential Helix Park context and why IoT is a suitable response. Second, it documents the technical design of the solution, including the sensing concept, the Wokwi hardware simulation, the Python artefact, and the data flow from generation to interpretation. Third, it evaluates how well the finished system performs against its intended purpose, while also reflecting on security, privacy, limitations, and realistic future development.

The final design combines an ESP32-style edge node with a Python desktop monitoring platform. The Wokwi node models one communal sensing point using a DHT22, an air-quality proxy input, a light sensor, and a vent switch. The Python artefact operates at building level by simulating five communal zones, applying safe/warning/danger rules, logging results, exporting charts, and presenting recommendations through a live dashboard. The scope of the solution is limited to communal environmental monitoring only; it does not include resident identification, apartment-level surveillance, or a real cloud back end.

The remainder of the report is structured as follows. Section 2 reviews the literature and industry guidance that informed the project. Section 3 explains the methodology and design choices. Section 4 discusses the implementation and how theory was translated into practice. Sections 5 and 6 present the results and critically evaluate them. The report then closes with a conclusion, future recommendations, references, and appendices.

2. Literature Review / Theoretical Framework

This section introduces the concepts and sources that influenced the project design. Because the artefact sits between smart buildings, environmental monitoring, and IoT security, it was important to draw on both academic literature and industry guidance rather than treating the solution as only a coding exercise.

2.1. Overview of Key Theories, Concepts, or Models

A useful starting point is the idea of layered IoT architecture. In practical terms, IoT systems are often separated into sensing, communication, processing, and application layers. That model is relevant here because the project deliberately distinguishes between the edge sensing node and the building-level dashboard. OASIS describes MQTT as a lightweight publish-subscribe messaging protocol designed for constrained devices and low-bandwidth environments, which makes it a natural reference point for smart-building telemetry even though the current project uses a simulated local data path rather than a live broker (Banks et al., 2019).

2.2. Critical Analysis of Relevant Literature

Research on indoor environmental quality monitoring also supports the use of multi-sensor observation rather than single-variable comfort tracking. Tsang et al. (2024) show that real-time indoor environmental quality monitoring becomes more useful when several readings are interpreted together, because occupant comfort and building performance are rarely explained by temperature alone. At the same time, US EPA guidance emphasises that indoor air quality is strongly influenced by ventilation, pollutant sources, and wider building management decisions, which supports the decision to include an air-quality proxy, light level, and vent state alongside temperature and humidity (EPA, 2025a; EPA, 2025b).

2.3. Identification of Gaps or Debates

The literature also raises some limits and tensions. Academic smart-building studies often assume richer hardware, calibrated sensors, or established infrastructure, while student artefacts have to remain practical and reproducible. There is also a gap between technically effective monitoring and ethically acceptable monitoring. In a residential context, even environmental sensing can become problematic if it is too closely linked to personal behaviour or if residents are unclear about what is being observed. This is why security and privacy guidance has to be considered alongside technical system design.

2.4. Justification for Chosen Approach

For security, the NISTIR 8259 and 8259A publications are especially useful because they frame IoT security in terms of device capability baselines, information flows, and manufacturer responsibilities rather than only abstract threats (Fagan et al., 2020a; Fagan et al., 2020b). In parallel, OWASP's IoT guidance highlights common weaknesses such as insecure interfaces, weak defaults, poor update mechanisms, and insufficient privacy protection (OWASP Foundation, n.d.). Taken together, the literature and industry guidance support the approach used in this project: a modular, privacy-aware, explainable IoT solution that prioritises transparent monitoring, manageable scope, and clear operational meaning over technical excess.

3. Methodology / Design

(Approximately 10–20% of total word count)

3.1. Research or Development Approach

The project followed a design-based development approach supported by simulation and iterative refinement. The goal was not to test a scientific hypothesis in a laboratory sense, but to create a technically credible IoT artefact that could be justified against the Helix Park scenario and evaluated through its outputs. This made a mixed design-and-evaluation approach more appropriate than a purely experimental one.

3.2. Methods Used

The main methods and tools used were Wokwi for hardware simulation, Python 3 for the software artefact, Tkinter for the GUI, Matplotlib for charts, JSON configuration files for repeatable thresholds and zone settings, CSV and JSON logging for output evidence, and unit tests for threshold, alert, and simulator validation. The system also used a deterministic seed so that scenario behaviour could be reproduced more reliably across runs.

Method / Tool	Use in the project	Reason for choice
Requirements-led design	Translated the Helix Residences brief into system requirements for monitoring, alerts, logging, visualisation, and recommendations.	Helped keep the artefact closely aligned to the assessment task.
Layered IoT architecture	Split the system into sensing, processing, alerting, logging, analytics, and GUI layers.	Reflected real IoT design practice and improved clarity.

Method / Tool	Use in the project	Reason for choice
Simulation-based development	Generated realistic environmental readings for communal zones instead of using physical building data.	Allowed a complete and testable system to be produced within the assessment constraints.
Zone-specific modelling	Assigned different patterns to the lobby, gym, co-working space, corridor, and rooftop lounge.	Made the data more believable and analytically useful.
Rule-based threshold analysis	Applied Safe, Warning, and Danger rules to temperature, humidity, air quality, light, and vent state.	Made decisions transparent and easy to justify in the report.
Scenario injection	Triggered faults such as ventilation failure, overheating, poor air quality, sensor fault, and stuck vent.	Helped test the system under abnormal operating conditions.
Python 3	Used as the main language for the full software artefact.	Met the brief and supported rapid development.

Method / Tool	Use in the project	Reason for choice
Tkinter GUI	Built the local dashboard, controls, status panels, and alert display.	Suitable for a simple desktop artefact that runs locally.
Matplotlib	Produced trend charts and rolling-average visualisations.	Supported data analysis and report evidence.
CSV and JSON logging	Stored readings and alerts as structured files.	Improved reproducibility and created an audit trail.
Wokwi hardware simulation	Modelled the ESP32 sensing node and basic actuator behaviour.	Represented the embedded IoT layer without physical hardware.
Deterministic testing	Used repeatable settings and unit tests for core logic.	Improved reliability and made the system easier to validate.
Privacy-by-design	Limited monitoring to communal environmental data only.	Supported ethical and data-minimisation requirements.

3.3. Justification of Choices

These methods were chosen because they balance technical realism with practical deliverability. Wokwi is appropriate for demonstrating the embedded sensing concept without requiring physical hardware. Python is suitable because the brief required a standard Python environment and because it allows the data pipeline, rules, analytics,

and interface to be explained clearly. Tkinter and Matplotlib are relatively lightweight but still strong enough for a report-ready desktop artefact. Compared with a full web stack or cloud back end, this combination reduced setup complexity while keeping the IoT design logic intact.

3.4. Ethical Considerations

Ethical considerations were built into the methodology from the start. The system monitors communal areas only, not private apartments. It does not use cameras, microphones, or identity-linked resident data. The data model was intentionally limited to environmental signals and zone-level operational interpretation. This is important because the brief specifically highlights privacy, consent, and data minimisation in residential settings.

3.5. Limitations of Methodology

The main methodological limitation is that the artefact is simulation-led rather than based on physically measured building data. The environmental readings are behaviourally modelled rather than calibrated against a live sensor network. The air-quality metric is also a proxy rather than a certified pollutant measure. These limitations were managed by keeping the thresholds transparent, making zone behaviour explainable, and ensuring that the results were still analytically useful rather than arbitrary.

4. Implementation / Analysis / Development

4.1. Overview

The implementation was organised so that each main IoT task had a clear place in the codebase. The simulator generates zone-specific environmental data, the rules engine classifies readings, the alert manager tracks threshold breaches and repeated issue patterns, the logger writes reproducible output files, the analytics layer creates summaries and charts, and the GUI presents the system to the user. This separation was important because it kept the artefact understandable and reduced the risk of turning the dashboard into a single large script.

4.2. Application of Theory into Practice

The ideas from the literature review were applied quite directly. The layered IoT concept shaped the distinction between edge sensing and central monitoring. MQTT was not implemented as a real transport, but it influenced how the data flow was described and justified. Indoor air quality literature and EPA guidance informed the choice to include air quality, ventilation, temperature, and humidity together instead of treating comfort as a one-variable problem. NIST and OWASP guidance informed the security discussion and the decision to keep the system narrow in scope and privacy-aware.

4.3. System Design, Models, or Processes

On the embedded side, the Wokwi node models one communal sensing point using an ESP32 DevKit, a DHT22, a light sensor, an air-quality proxy input, a slide switch representing vent state, and LEDs plus a buzzer for local response. The ESP32 classifies safe, warning, and danger states locally. On the monitoring side, the Python dashboard acts as a building-level interface. It collects simulated zone readings, displays live tables, highlights statuses, presents selected-zone analytics, and exports charts. This gives the project a recognisable edge-to-dashboard architecture rather than only a standalone desktop simulation.

4.4. Logical Progression of Steps

Three short code extracts are included below because they show how the artefact moves from simulated sensor values to decisions that a building manager could actually use. I kept these examples brief so that they support the explanation rather than turning the report into a code listing.

5. Selected Code Extracts

The first extract comes from the rule engine. It classifies every metric separately and then promotes the zone to Safe, Warning, or Danger depending on the worst active condition. This was important because a single poor condition, such as low ventilation or bad air quality, should not be hidden by otherwise normal readings.

```
def classify_metric_states(self, reading: SensorReading) -> Dict[str, str]:
    return {
        "temperature": self._range_status(reading.temperature,
self.thresholds["temperature"]),
        "humidity": self._range_status(reading.humidity,
self.thresholds["humidity"]),
        "air_quality": self._max_only_status(reading.air_quality,
self.thresholds["air_quality"]),
        "light_level": self._min_only_status(reading.light_level,
self.thresholds["light_level"]),
        "vent_open_ratio": self._range_status(reading.vent_open_ratio,
self.thresholds["vent_open_ratio"]),
    }
```

The second extract comes from the simulator. This section is where the data stops looking random and starts behaving like a building. Different communal spaces respond differently to occupancy and time of day, and the anomaly scenarios push the readings in believable directions rather than changing everything at once.

```
if zone.name == "Gym":
    values["temperature"] += occupancy * 1.3
    values["humidity"] += occupancy * 5.0
elif zone.name == "Co-working Space":
    values["air_quality"] += occupancy * 140

if scenario == "ventilation_failure":
    values["air_quality"] += 420
    values["humidity"] += 8
    values["vent_open_ratio"] = 0.02
```

The third extract comes from the alert manager. I wanted the system to do more than react to single bad samples, so repeated non-safe states also create a maintenance-oriented alert. This supports the facilities management aim of the brief because it highlights patterns that may justify an inspection.

```
if len(history) == self.repeated_warning_count and all(status != "Safe" for status
in history):
    created.append(
        self.create_alert(
            assessment,
            "Warning" if assessment.status == "Warning" else assessment.status,
            f"{assessment.zone}: repeated non-safe environmental pattern detected.",
            f"Schedule a facilities inspection for {assessment.zone} due to repeated
comfort violations.",
            "repeated_pattern",
            scenario,
        )
    )
```

The main implementation sequence was as follows. First, the zone configuration and threshold files were created so that the system had a stable basis for simulation. Second, the simulator was designed to produce distinct patterns for each communal space. Third, the rule engine and alerts were added so that the data had operational meaning. Fourth, logging and chart export were added to make the outputs useful in

the report. Finally, the GUI brought the whole system together and exposed the features through controls for zone selection, metric selection, simulation speed, and anomaly injection.

6. Results / Findings

6.1. Presentation of Outcomes

The final system ran as a local desktop application and produced live readings across five communal zones: Lobby, Gym, Co-working Space, Corridor, and Rooftop Lounge. In the current evidence set, the system generated 260 recorded readings and 328 alerts across the saved test runs. Each zone contributed 52 readings, which shows that the simulation and logging pipeline handled the monitoring cycle consistently across the building model.

6.2. Data Visualisation

Figure 3 shows the full desktop dashboard exactly as it appears during live monitoring. This figure is useful because it brings the main software functions together in one view rather than separating them into isolated screenshots.

At the top of the interface, the summary cards give a fast building-level picture by showing overall status, the number of active alerts, the current worst-performing zone, and the selected scenario. This means a facilities user does not need to scan the whole table before understanding whether the building is stable or whether attention is needed.

The middle section combines the simulation controls, a live monitoring table, and selected-zone snapshot cards. The controls are there to start, pause, reset, change speed, switch zones, and apply anomaly scenarios. The table then shows the live readings for each communal area, while the snapshot cards make the selected zone easier to read during a demonstration.

On the right-hand side, the selected-zone overview, operational recommendations, and recent alerts panels turn the raw readings into short operational messages. At the bottom, the trend chart and analytics summary help the user interpret whether the current issue is isolated, stable, or part of a repeated pattern over time.

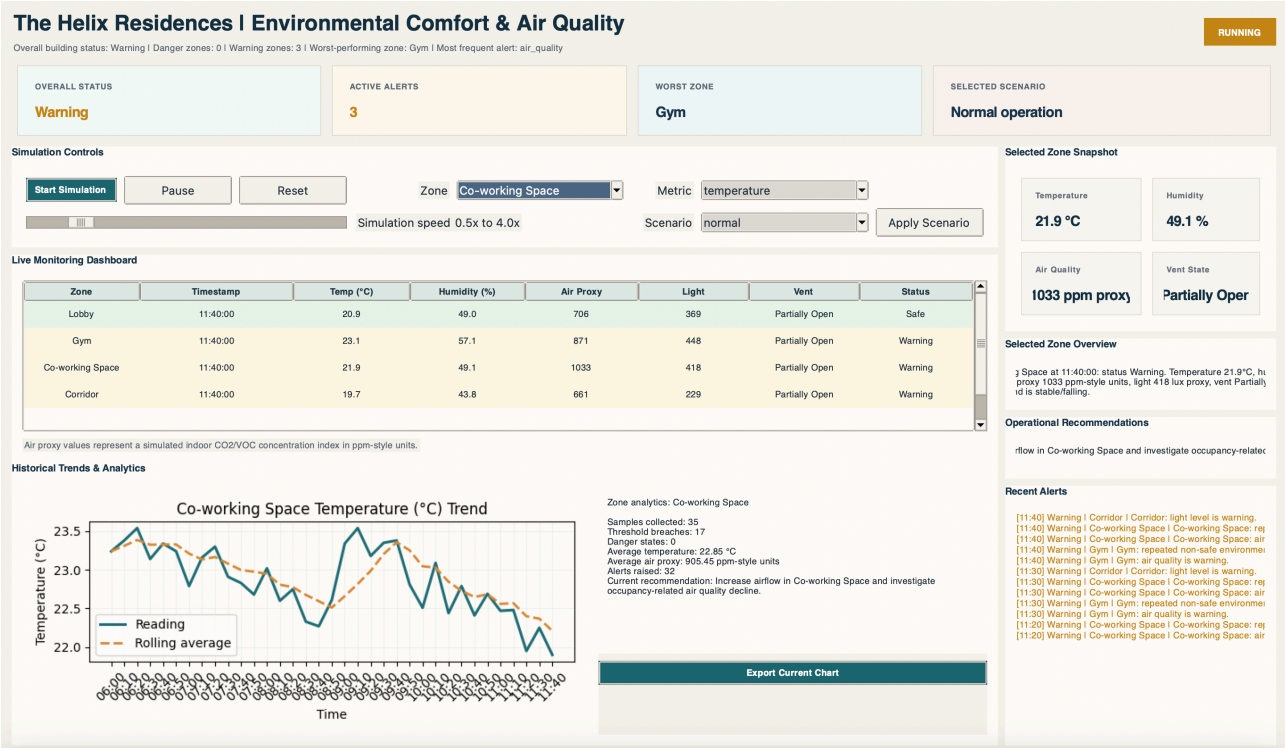

Figure 3: Full Python monitoring dashboard showing summary cards, live readings, control inputs, charted history, recommendations, and recent alerts.

The recorded outputs show that the Gym and Co-working Space generated the highest average air-quality proxy values, at approximately 1079 and 1044 ppm-style units respectively, while the Corridor and Rooftop Lounge remained lower. Alert history was dominated by air-quality alerts and repeated-pattern warnings, showing that the system was not only reacting to sudden threshold breaches but also identifying persistent non-safe conditions over time. These results are visually supported by the exported trend charts and by the zone-level analytics in the dashboard.

6.3. Objective Reporting

The results also show the effect of anomaly scenarios. Ventilation failure produced the largest cluster of alert activity, while poor air quality and sensor fault scenarios also generated observable changes in the logs. In the stored outputs, 122 alerts were recorded as danger and 206 as warning. This suggests that the system can represent a useful spread of normal, degraded, and clearly problematic conditions rather than collapsing into only one state type.

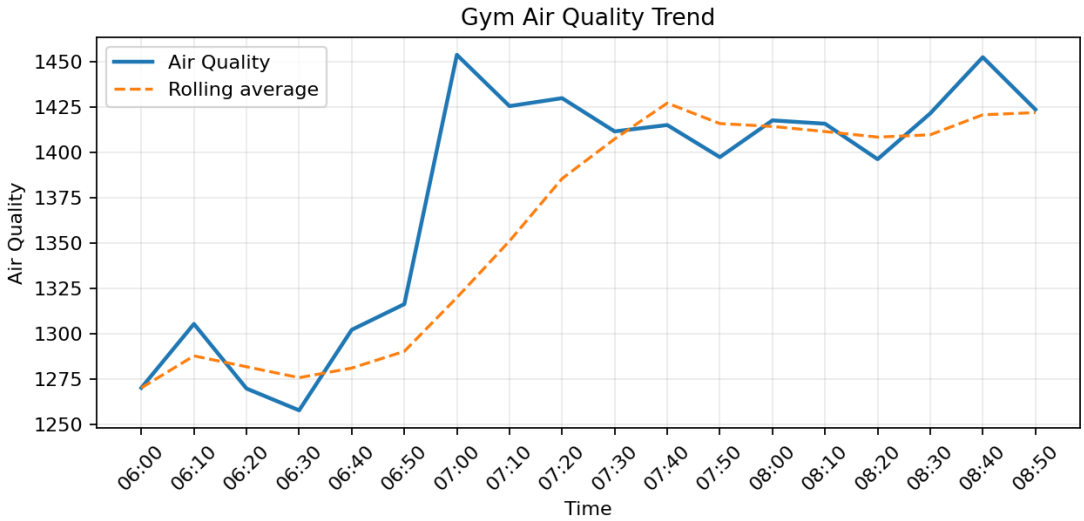

Figure 1: Exported gym air-quality trend from the Python monitoring dashboard.

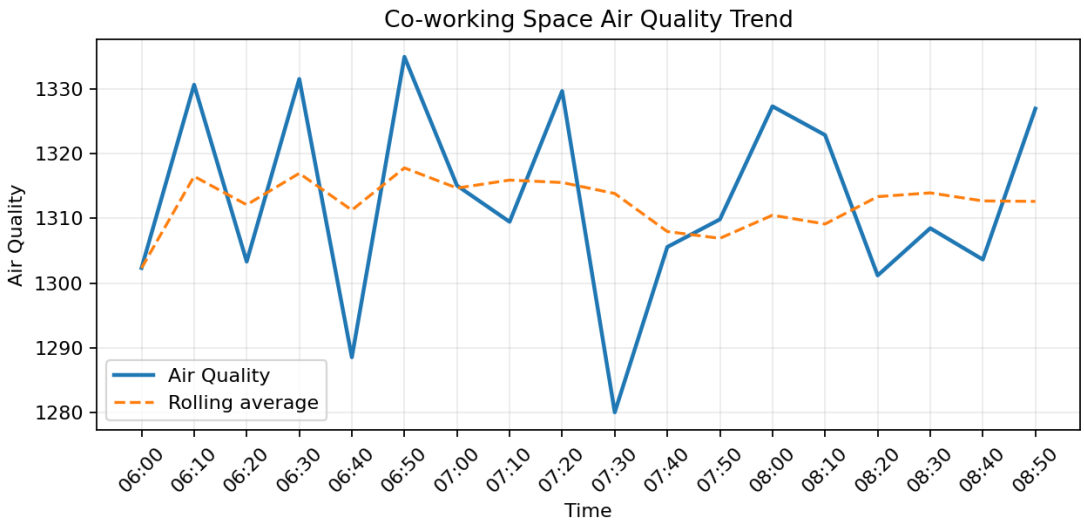

Figure 2: Exported co-working space air-quality trend showing sustained decline behaviour.

7. Discussion / Critical Evaluation

7.1. Interpretation of Findings

The findings suggest that the system succeeds in turning simulated environmental data into operationally useful information. The most important point is not the exact numerical value of any single reading, but the way the dashboard connects those values to zone-level interpretation. For example, the Gym becoming the worst-performing zone under ventilation-related conditions is not surprising in itself, but the

dashboard makes that pattern visible quickly and connects it to a sensible recommendation for facilities staff.

7.2. Evaluation Against Objectives

Against the original aims, the project largely achieved what it set out to do. It produced a working desktop artefact, simulated multiple communal zones, generated realistic enough environmental patterns to support discussion, logged outputs reproducibly, and included a clear privacy-aware scope. The objectives that remain only partially met relate to real-world deployment rather than the artefact itself, such as live communications infrastructure, production security controls, and validated building standards integration.

7.3. Comparison with Literature or Expected Outcomes

Compared with the literature, the project aligns reasonably well with the idea that indoor environmental quality monitoring should be multi-factor and connected to building operations rather than isolated data display. That is consistent with Tsang et al. (2024), who emphasise real-time indoor environmental quality monitoring using multiple signals, and with EPA guidance that links indoor conditions closely to ventilation and pollutant management (EPA, 2025a; EPA, 2025b). Where the project differs from richer research systems is in its deliberate simplicity: it uses a proxy air-quality measure, a local dashboard, and simulated zone behaviour rather than a full deployed sensing network.

7.4. Strengths and Weaknesses

In terms of strengths, the project is strongest in its modularity, explainability, and ability to generate evidence. The dashboard is clear enough to demonstrate the system live, the logs are useful for later analysis, and the zone-specific behaviour makes the results more believable. The main weaknesses are the reliance on simulation, the limited hardware realism of the Wokwi node, and the fact that the current deployment model is still single-machine and local. These are acceptable for an academic artefact, but they would matter much more in practice.

7.5. Consideration of Reliability and Validity

Reliability and validity need to be considered carefully. Reliability is helped by the deterministic seed, structured logging, and repeatable threshold logic. Validity is more limited because the system is not drawing on calibrated real building data, and the air-

quality measure is only a proxy. As a result, confidence in the exact numbers should be modest, but confidence in the overall behaviour, architecture, and operational interpretation is stronger. In other words, the project is more valid as a simulation of IoT monitoring logic than as a direct predictor of a real building's exact environmental conditions.

The project also shows the value of combining technical structure with communication clarity. A monitoring platform is only useful if its outputs can be understood quickly by the people expected to act on them. The dashboard, alert history, exported charts, and recommendation panel all contribute to that goal. This is one of the reasons I think the artefact fits the higher assessment band: it does not stop at collecting data, but instead tries to connect design, implementation, evidence, and action in one coherent system.

A further point from the evaluation is that the project performs best when it is treated as a decision-support system rather than as a precise environmental measurement instrument. That distinction matters. The artefact is strong at showing how zone behaviour can be monitored, classified, and escalated, but it is deliberately less strong at claiming laboratory-style measurement accuracy. In my view, that is still acceptable because the purpose of the project is to demonstrate IoT system design, operational interpretation, and critical judgement rather than to replace a professionally commissioned building monitoring installation.

8. Conclusion

This report has presented a simulated IoT monitoring system for The Helix Residences, focusing on environmental comfort and air quality in communal spaces. The project addressed a clear operational problem, justified the use of IoT in a residential smart-building context, and implemented a working artefact that links sensing, classification, analytics, logging, and dashboard presentation. Overall, the system meets its main objectives as a university artefact and demonstrates a strong understanding of IoT architecture, privacy-aware design, and operational decision support. Its main limitations come from the fact that it remains a simulation rather than a fully deployed building system, but within that scope it provides a coherent and well-structured solution.

9. Recommendations / Future Work

9.1. Practical Improvements

The most practical improvement would be to connect the edge sensing node to the dashboard through a real message path, such as MQTT, and to store data in a dedicated database rather than local files. A second useful improvement would be to refine the comfort thresholds against published building guidance and to expose those thresholds more directly in the interface for auditing and tuning.

9.2. Further Research Directions

Further investigation would be useful in three areas. First, a future version could test whether the same dashboard logic still works well when data comes from multiple live nodes rather than a simulator. Second, the relationship between ventilation state and indoor air proxy values could be modelled more realistically. Third, the project could be extended to compare alerting strategies, for example simple thresholds versus moving baselines or lightweight prediction.

9.3. Scalability and Real-World Application

From a real-world perspective, the system could scale into a building-wide or block-wide deployment if each communal area used a similar edge node and published readings into a central broker. That would make the dashboard a more genuine smart-building operations tool and would also make integration with a wider Helix Park data layer more realistic. However, scaling would only be responsible if it happened alongside stronger device authentication, encrypted communications, access control, and resident transparency measures.

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